MILITARY SPECIFICATION

CONTAINER, QUADRUPLE (QUADCON)

This specification is approved for use within the United States Marine Corps, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the manufacture and acceptance of the quadruple container (QUADCON). The QUADCONs are designed to be shipped in sets of four in the configuration of an International Organization for Standardization (ISO) 20 foot (ft) container. A container rack which provides support for 36 insert containers is required when the QUADCON is used in its optional bin/drawer configuration.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issues of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commandant of the Marine Corps, Code LMA-2, Washington, D.C. 20380, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8145

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SPECIFICATIONS:

MILITARY

MIL-T-704	Treatment and Painting of Materiel
MIL-C-5541	Chemical Conversion Coatings on Aluminum Alloys
MIL-Q-9858	Quality Program Requirements
MIL-C-29446	Container, Insert (Insert)
MIL-C-29448	Container, Rack, Quadruple (QUADCON Rack)

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-209	Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment
MIL-STD-810	Environmental Tests Methods and Engineering Guidelines
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking and Waterproofing, with Appropriate Test Methods
MIL-STD-1791	Designing for Internal Aerial Delivery in Fixed Wing Aircraft

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

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DRAWINGS

MARINE CORPS

835040A0000	Container, Quadruple (QUADCON)
835042A0000	Container Rack, Quadruple (QUADCON Rack)
835043A0000	Container, Insert (INSERT)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be in the issue of the nongovernment documents which is current on the date of the solicitation.

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 668	Freight Containers--External Dimensions and Ratings
ISO 1161	Series I Freight Containers--Corner Fittings--Specifications
ISO 1496/I	Series I Freight Containers--Specification and Testing--Part I: General Cargo Containers
ISO 1496/VII	Series I Freight Containers--Specification and Testing--Part VII: Air Mode Containers

(Application for copies should be addressed to International Organization for Standardization, P.O. Box 56, 1211 Geneva 20, Switzerland.)

U.S. DEPARTMENT OF COMMERCE

PS-1	Softwood Plywood, Construction and Industrial

(Application for copies should be addressed to the Superintendent of documents, Government Printing Office, Washington, D.C. 20402.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1. First article. When specified in the contract or purchase order a sample shall be subject to first article inspection (see 4.4 and 6.3). The first articles shall be manufactured using the same procedures and tooling proposed for production. Approval of the first articles by the acquiring activity shall not relieve the Contractor of the obligation to supply equipment conforming to this specification. Changes or deviations in production units for the approved first article baseline shall require written authorization by the acquiring activity.

3.2 Materials. Materials and components shall conform to the requirements specified herein and in Drawing Numbers (No.) 835040A0000, 835042A0000, 835043A0000, and 945008A0030.

3.2.1 Plywood. The plywood panel core material shall conform to Department of Commerce PS-1, grade A-A (plugged) or better, bonded with exterior glue. All plywood for the floor, roof, and walls shall be treated in accordance with Drawing No. 835040A0000 to prevent moisture absorption and rotting.

3.2.2 Fiberglass reinforced plastic. All panel core material shall be laminated on both surfaces with 18 ounce per square yard (+10 percent) woven roving fiberglass cloth reinforcement using thermosetting polyester resin. The surface laminates shall contain not less than 42 percent glass and a maximum of 58 percent polyester resin by weight. The woven roving cloth shall be continuous in both directions with no ties or laps. The exterior panel surfaces shall be coated with a gel coat fused under heat and pressure. The gel coat shall be ultraviolet; weather, mildew, oil, and chemical resistant; and shall conform to the color and finish requirements of 3.9.

3.2.3 Dissimilar materials. Intimate contact of dissimilar materials, as defined in MIL-STD-889, which can be expected to cause galvanic corrosion, shall be avoided. When such contact cannot be avoided, an interposing insulating material shall be provided to minimize the corrosive effect.

3.3. Design and construction

3.3.1 Understructure. The understructure shall be fabricated in accordance with Drawing No. 835040A0000. All cross members

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shall be of sufficient strength to provide uniform floor support on all loading conditions, when supporting solely by the upper and lower corner fittings, and when being lifted/transported utilizing the forklift pockets. The maximum recess allowed for the bottom side rails and end frame rails above the plane of the bottom corner fittings shall be 0.50 in (13 mm). The underside shall be flat and flush with the bottom faces of the lower corner fittings as permitted for 1F containers in Section 4.3.2 of ISO 1496/I. No part of the base shall deflect more than 0.50 in (13 mm) below the bottom faces of the lower corner fittings when supported by the corner fittings and loaded as prescribed in Section 4.3.3 of ISO 1496/I. This applies to the 1F configuration in which case the value of R is 11,200 pounds (lbs) (49,840 Newtons). Testing shall be in accordance with 4.7.1.

3.3.2 Roof. The roof panels, roof rails, and upper frame members shall be at least 0.25 in. (6 mm) below the top plane of the upper corner fitting.

3.3.3 Doors. Double doors shall be provided on each 57.5 in (1,461 mm) side. These doors shall be interchangeable with like doors and shall be fabricated in accordance with Drawing No. 835040A0000. The doors shall be capable of withstanding without damage a static load of 200 lbs. (890 Newtons) acting on the edge opposite the hinge pivot, a dynamic load on the door of 50 lbs (222 Newtons) applied to the midpoint of the locking edge acting in a horizontal plane perpendicular to the front panel of the door in both directions, and the force increased by dropping the weight (50 lbs) a distance of 12 in (305 mm). Testing shall be in accordance with 4.7.2.

3.3.4 Container rack, QUADCON. The QUADCON shall be capable of housing a QUADCON container rack built in accordance with MIL-C-29448(MC). When the rack is fully erected within the QUADCON, 36 insert containers built in accordance with MIL-C-29946(MC) shall fit within the QUADCON with the QUADCON doors closed. Testing shall be in accordance with 4.7.3.

3.3.5 Interchangeability. Like demountable hardware and components of the QUADCON shall be interchangeable.

3.3.6 Coast Guard certification. The QUADCON shall be certified as an 82 in by 57.5 in. by 96 in. (2,083 mm by 1,461 mm by 2,438 mm) 1F ISO container for its maximum rated load in accordance with the applicable Coast Guard regulation for such containers for all types of transport, including motor carrier (truck), sea, railroad, and air. Qualification shall be in accordance with the requirements of ISO 1496/I. At a minimum, testing shall be in accordance with 4.7.4 through 4.7.9.

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3.3.7 Metal fabrication. Metal used in the fabrication of equipment shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and punching shall be done neatly and accurately. Corners shall be square and true. All bends shall be made with controlled means in order to ensure uniformity of size and shape.

3.3.8 Facing material. Panel facings shall be fiberglass coated plywood, smooth or embossed, and shall be free from bubbles, glazing, cracks, and discontinuities.

3.3.9 Bolted and riveted connectors. Bolt and rivet holes shall be accurately punched or drilled and have the burrs removed. Washers, lockwashers, or locknuts shall be provided where necessary and all bolts, nuts, and screws shall be tight. Rivet heads, when not countersunk or flattened, shall be of uniform size and shape for the same diameter rivet, concentric with the rivet holes, and in full contact with the surface of the members.

3.3.10 Welding. The surface of the parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Weld penetration shall be such as to provide transference of maximum design stress through the base metal juncture. Weld configuration and sizes shall be as specified on the applicable drawings.

3.3.11 Castings. Casting shall be sound and free from patching, misplaced coring, warping, or defects which might render the casting unsound for use.

3.3.12 Seals. Seals shall be installed so that the fit is snug and continuous contact is maintained with mating seal faces. Mitered end junctions of vertical and horizontal seal runs shall be accurate, and gaps shall be sealed as specified on the applicable drawings.

3.4 Maintainability. The QUADCON has been designed so that as much of the maintenance, as possible, may be performed as far forward (preferably by the using unit), as feasible.

3.4.1 Organizational level maintenance. Preventive maintenance (PM) consists primarily of inspecting for cleanliness and mechanical integrity, cleaning, lubricating hinges and latches as required. PM is performed by designated Marines in the supply section at the smallest unit level. Corrective maintenance, such as replacing minor parts and repainting is performed by the unit's organic maintenance personnel. No specific MOS has been designated for second echelon maintenance; only common tools in maintenance work spaces are required.

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3.4.2 Intermediate Level Maintenance. Intermediate level maintenance (3d and 4th echelons) is performed by the Maintenance Battalion, FSSG. Maintenance will include patching to maintain weatherproofness, testing the alignment of structual members and realigning as required, replacing major parts such as QUADCON corner fittings, replacing contain door weatherstripping; and performing modification as directed. Typical MOS could be 1316 (metalworker), 1341 (Engineer equipment mechanic), 2161 (machinist), and 2871 (TMDE technician) or 2874 (metrology technician).

3.4.3 Depot Level Maintenance. Depot level maintenance will be accomplished on a regional bases using organic or contract services.

3.5 Transportability.

3.5.1 Stacking. The QUADCON shall withstand the requirements of the Coast Guard intermodal transportation requirements for land, sea, and air transport as 1C shipping containers when four QUADCONs are coupled together with a gross weight of 11,200 lbs. (49,840 Newtons). Testing shall be in accordance with 4.7.13.7.

3.5.2 Railroad transportability. The QUADCON shall withstand without damage the shocks normally induced by transport by the railroads and the shocks induced by MIL-STD-810, Method 516.3, Procedure VIII, with the payload uniformly distributed over the floor. The impact speed shall be between 9 miles per hour (mph) (14.5 kilometers per hour (km/hr)) and 10 mph (16.1 km/hr) in both directions of the longitudinal axis (2 impacts in each direction). This test shall be done with the container rack and inserts inside if specified (see 6.2). Testing shall be in accordance with 4.7.15.1.

3.5.3 Air Transportability. The QUADCON shall be capable of withstanding rapid decompression in accordance with MIL-STD-1791 when transported aboard cargo aircraft. The QUADCON shall be so configured for air shipment as to prevent any part from becoming a projectile in the event of catastrophic loss of aircraft cabin pressure. If a pressure relief or breathing device is installed, the QUADCON shall retain its weatherproofness. This test shall be done with the container racks and inserts inside, if specified (see 6.2). Testing shall be in accordance with 4.7.13 and 4.7.15.2.

3.6 Environmental requirements.

3.6.1 Weatherproofness. The QUADCON shall withstand testing as described in paragraph 4.7.13.1 without leakage at any location.

3.6.2 Temperature-humidity. The QUADCON shall be capable of functioning and withstanding exposure to temperatures of -65°F to 125°F at any possible relative humidity from 0 to 100 percent. In storage, the QUADCON shall be capable of withstanding exposure to temperatures of -70°F to 160°F at any possible relative humidity from 0 to 100 percent without cracking, corroding, delamination, or peeling of paint. Testing shall be in accordance with 4.7.13.2 and 4.7.13.3.

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3.6.2.1 Thermal shock. The QUADCON shall be capable of withstanding sudden temperature changes up to and including from 160°F to -70°F to 160°F without degradation of physical properties. Testing shall be in accordance with 4.7.13.4.

3.6.3 Snow and ice. The QUADCON shall be capable of functioning in accordance with the requirements specified herein during periods of snow and ice storms. The QUADCON shall inhibit the collection of snow and ice such that its primary function is not degraded. The QUADCON shall be capable of withstanding snow loads up to 40 lbs per square foot (1,915 N/m) in addition to its internal loads.

3.6.4 Wind. The QUADCON shall be capable of withstanding wind speeds of up to 65 mph (104 km/hr) and gusts of up to 120 mph (192 km/hr). In winds in excess of 55 mph (88 km/hr), ground anchors are required.

3.6.5. Roof load. The QUADCON shall withstand a uniform load of 40 lbs per square foot (1,915 N/m) over the entire roof of the QUADCON without permanent damage and a concentrated load of 660 lbs (2,936 Newtons) over an area 24 in by 12 in (610 mm by 305 mm) without permanent damage rendering the QUADCON unusable. Testing shall be in accordance with 4.7.13.5 and 4.7.13.6.

3.6.6 Sand and dust. The QUADCON shall withstand the effects of sand and dust on all external moving parts without degradation or infiltration of particles inside the unit. Testing shall be in accordance with 4.7.13.8.

3.6.7 Sunshine. The QUADCON shall withstand the effects of ultraviolet radiation (sunshine) without degradation or effect on serviceability of the QUADCON's components or materials during the service life, and shall withstand the test requirements of MIL-STD-810, Method 505.1. Testing shall be in accordance with 4.7.13.9.

3.6.8 Solar load. The QUADCON shall withstand a solar heat load in its roof sufficient to raise the outer skin to at least 200°F (93.3°C) without any evidence of delamination or permanent deformation. Testing shall be in accordance with 4.7.13.10.

3.6.9 Marine environment. The QUADCON shall be fully serviceable when exposed to sea-salt fallout equivalent to 25 pounds/acre/year (2.8 grams/m^2/year). All fasteners, seals, or other hardware shall show no evidence of corrosion or degradation affecting serviceability when tested as required by 4.7.13.11.

3.6.10 Drop shock.

3.6.10.1 Flat drop. The QUADCON shall withstand a flat drop of 12 in (305 mm) from the ground onto a concrete surface. Testing shall be in accordance with 4.7.14.1.

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3.6.10.2 Rotational drop. The QUADCON shall withstand rotational drops with one edge placed on a 4 in. (102 mm) block and the opposite edge raised 12 in. (305 mm) and allowed to free fall onto a concrete surface with the 4 in. (102 mm) block acting as a pivot. The QUADCON shall withstand this drop on each bottom edge. Testing shall be in accordance with 4.7.14.2.

3.7 Dimensions. The exterior envelope of the QUADCON shall comply with the ISO Shipping Container Standards as defined in ISO 1946/I for a 57.5 in. (1,461 mm) long 1F containers. When coupled together, the four container array shall be capable of being handled and shipped as a 1C container, except the height shall be limited to 82 in. (2,083 mm). The applicable drawings fully describe all pertinent dimensions and tolerances. Should any dimensions and tolerances result in a component or assembly which will not meet the requirements of this specification, the supplier is responsible for fabrication techniques which will prevent the occurrence of such dimensions and tolerances. Testing shall be in accordance with 4.7.11.

3.8 Weight. The tare weight of the QUADCON, including all permanent accessories, shall be 2,010 lbs. ± 5 percent. The gross weight shall not exceed 11,200 lbs. The actual tare weight shall be stamped on the nameplate. Testing shall be in accordance with 4.7.12.

3.9 Finish.

3.9.1 Surface preparations. All surfaces shall be prepared as specified on the applicable drawings. All aluminum alloy materials shall be protected by a chemical film conforming to MIL-C-5541 even if a final finish is not applied.

3.9.2 Priming and final finish. All surfaces to be painted shall be cleaned and treated in accordance with MIL-T-704 (Table II), type B, except that all surfaces or ferrous components shall be coated with a primer coat and topcoat in accordance with the applicable drawings.

3.10 Identification marking.

3.10.1 Identification. Each QUADCON shall have the following information provided on a serial plate permanently affixed to the QUADCON in accordance with MIL-STD-130:

Manufacturer:

Model:

Type:

Serial No.: (As assigned by the Contracting Activity)

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Tare Weight:

NSN:

3.10.2 Markings. Markings shall be in accordance with Drawing No. 835040A0000 and in compliance with coding, marking, and Convention for Safe Containers (CSC) plating requirements.

3.10.3 Certification. Each QUADCON shall carry the appropriate certification (CSC safety approval plate) from the approval authority as to its capability to act as a shipping container.

3.11 Workmanship. All parts, components, and assemblies of the QUADCON, including castings, forgings, molded parts, stampings, seals and sealing agents, machined surfaces, and welded parts, shall be cleaned and free from any defects that will reduce the capability of the QUADCON to meet the requirements specified herein. Any components and assemblies which have been repaired or modified to overcome deficiencies or deficient components or assemblies which are determined to be used "as is" shall not be used unless authorized by the contracting activity. External surfaces shall be free from burrs, slag, sharp edges and corners, except where sharp edges and corner are required.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the Contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the Contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set for in this specification shall become a part of the Contractor's overall inspection system or quality program in accordance with MIL-Q-9858. The absence of any inspection requirements in this specification shall not relieve the Contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.5)

c. Individual inspection (see 4.6)

4.3. Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.4, 4.5, and 4.6.

4.4. First article inspection. Prior to beginning production of the QUADCON, the Contractor shall fabricate four QUADCONs and subject two of them to the tests specified in Table I, except when four QUADCONs are required.

4.5. Quality conformance inspection. From each shipping lot, a single QUADCON shall be chosen at random by the Government and subjected to the tests specified in Table I. Failure of any test shall be cause for inspection of the entire lot at the contractor's expense for similar defects. Defective QUADCONs shall not be delivered without correction of the deficiency and then only with permission of the contracting officer.

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TABLE I. Tests

INSPECTION	REQ PARA	TEST PARA	1ST ART	QUAL CONF	IND
Lift by fork pockets	3.3.1	4.7.1	X		
Door Load	3.3.3	4.7.2	X		
Compatibility	3.3.4	4.7.3.1	X		
Compatibility	3.3.5	4.7.3.2	X		
Transverse and longitudinal rigidity	3.3.7	4.7.4	X		
Lifting from top	3.3.7	4.7.5	X		
Lifting from bottom	3.3.7	4.7.6	X		
Endwall strength	3.3.7	4.7.7	X		
Sidewall strength	3.3.7	4.7.8	X		
Longitudinal restraint	3.3.7	4.7.9	X		
Coast Guard certification	3.3.7	4.7.10	X		
Dimensional	3.7	4.7.11	X	X	X
Weight	3.8	4.7.12	X	X	X
Weatherproofness	3.6.1	4.7.13.1	X	X	
Humidity resistance	3.6.2	4.7.13.2	X		
Temperature range	3.6.2	4.7.13.3	X		
Temperature shock	3.6.2.1	4.7.13.4	X		
Roof load, uniform	3.6.5	4.7.13.5	X		
Roof load, concentrated	3.6.5	4.7.13.6	X		
Stacking	3.5.1	4.7.13.7	X		
Sand and dust	3.6.6	4.7.13.8	X		
Sunshine	3.6.7	4.7.13.9	X		
Solar load	3.6.8	4.7.13.10	X		
Marine atmosphere	3.6.9	4.7.13.11	X		
Flat Drop 1/	3.6.10.1	4.7.14.1	X	X	
Rotational drop 1/	3.6.10.2	4.7.14.2	X	X	
Railroad transportability 1/	3.5.2	4.7.15.1	X		
Air transportability 1/	3.5.3	4.7.15.2	X		

1/ With rack and inserts if specified (see 3.5.2, 3.5.3, and 3.6.10).

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4.6 Individual inspection. Each QUADCON shall be subjected to the inspections and tests specified in Table I. Failure to meet the requirements specified herein shall be cause for rejection. No QUADCON shall be reworked and retested without the expressed permission of the contracting activity.

4.7 Tests.

4.7.1 Lift by fork pockets. The QUADCON to be tested shall be uniformly loaded to a gross weight of 11,200 lbs. (49,818 Newtons). The QUADCON shall then be supported by two horizontal bars, each 8 in. (20 mm) wide, all the way through the fork pockets on one side of the QUADCON with sufficient length protruding outside the walls (doors) to attach lifting cables. The QUADCON shall then be lifted clear of the ground using only the bars for a period of 5 minutes and then lowered to the ground. The procedure shall then be repeated using the set of fork pockets 90° from those already tested. Upon completion of this test, the QUADCON shall show neither permanent deformation nor abnormality which will render it unsuitable for use and the dimensional requirements of 4.7.11 shall then be satisfied.

4.7.2 Door load. Each door of the QUADCON under test shall have a vertical force of 200 lbs. (890 Newtons) applied at the edge opposite from the hinge pivot with the door open at 90°. After 30 minutes the force shall be removed and the door examined for evidence of unbonded components, damage to hardware, and improper door operation. The door shall then be open to its maximum, a fixture shall be attached to the midpoint of the locking edge of the door which shall permit the application of 50 lbs. (222 Newtons) using free running cables to transmit the forces from the weight and 1/4 diameter steel cable. The weight of 50 lbs. (222 Newtons) shall be dropped five times from a height of 12 in. (305 mm). This shall be repeated from both sides of the door and shall be repeated on all four doors of the QUADCON. Any evidence of deformation, damage, or improper door operation shall constitute failure of this test.

4.7.3 Compatibility.

4.7.3.1 Container rack with inserts. A container rack built in accordance with MIL-C-TBD and one or more container inserts built in accordance with MIL-C-TBD shall be required for this test. Assemble the rack into the QUADCON and assure compatibility of 36 insert locations by installing the insert container into each location.

4.7.4 Transverse and longitudinal rigidity. Four QUADCONs shall be coupled together to form a 1C container and placed on four level supports, one under each extreme end corner fitting, and shall be

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restrained against lateral and vertical movement by means of anchor devices, acting through the bottom corner fittings. Lateral restraint shall be provided only at a bottom corner fitting diagonally opposite to and in the same end frame as a top corner fitting to which force is applied. When testing two end frames separately, vertical restraint shall be applied only at the end frame under test. A force of 33,600 lbs. (149,460 Newtons) shall be applied separately to each of the top corner fittings on one side of the QUADCONs under test in lines parallel to both the base and to the planes of the ends. The force shall be applied first towards, then away from, the top corner fittings. Upon completion of this test, each QUADCON, and the assembly, shall show neither permanent deformation nor abnormality which will render it unsuitable for use, and shall meet the dimensional requirements of 4.7.11. This test shall be repeated with the forces of 16,800 lbs. (74,730 Newtons) acting parallel to the planes of the sides of the QUADCONs.

4.7.5 Lifting from top. The four QUADCONs under test shall be coupled together to form a 1C container and shall then have a load uniformly distributed over the floor in such a way that the combined weight of the four QUADCONs and the test load is equal to 89,600 lbs. (398,540 Newtons). The QUADCON assembly shall then be carefully lifted from the top corner fittings at the ends of the assembly (four only) in such a way that no significant acceleration or deceleration forces are applied. The QUADCON assembly shall be suspended by the lifting mechanism for 5 minutes and then carefully lowered to the ground. Upon completion of this test, each QUADCON, and the assembly, shall show neither permanent deformation nor abnormality which will render it unsuitable for use and the dimensional requirements of 4.7.11 shall then be satisfied for each QUADCON individually and as an assembly.

4.7.6 Lifting from bottom. The four QUADCONs under test shall be coupled together to form a 1C container and shall then have a load uniformaly distributed over the floor in such a way that the combined weight of the four QUADCONs and the test load is equal to 89,600 lbs. (398,540 Newtons). They shall then be carefully lifted from the side apertures of the four bottom corner fittings at the ends of the assembly in such a way that no significant acceleration or deceleration forces are applied. Lifting forces shall be applied at 45° from the vertical, parallel to the long sides of the QUADCON assembly. In each case, the line of action of the lifting force and the outer face of the corner fitting shall be no further apart than 1.5 in. (38 mm). The lifting shall be carried out in such a manner that the lifting devices bear on the bottom corner fittings only. The QUADCON assembly shall be suspended for 5 minutes and then lowered carefully to the ground. Upon completion of this test, each QUADCON, and the assembly shall show neither permanent deformation nor abnormality which will render it unsuitable for use and the dimensional requirements of 4.7.11 shall then be satisfied for each QUADCON individually and as an assembly.

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4.7.7 Endwall strength. One endwall of the QUADCON under test shall be loaded to 6,947 lbs. (30,900 Newtons), uniformly distributed over the entire endwall surface. The endwall shall be supported to allow free deflection of the wall. Upon completion of this test, the QUADCON shall show neither permanent deformation nor abnormality which will render it unsuitable for use and the dimensional requirements of 4.7.13 shall then be satisfied for a single QUADCON.

4.7.8 Sidewall strength. One sidewall of each QUADCON under test shall be loaded to 6,947 lbs. (30,900 Newtons), uniformaly distributed over the entire sidewalls surface. The sidewall shall be supported to allow free deflection of the wall. Upon completion of this test, the QUADCON shall show neither permanent deformation nor abnormality which will render it unsuitable for use and the dimensional requirements of 4.7.11 shall then be satisfied for a single QUADCON.

4.7.9 Longitudinal restraint. The four QUADCONs under test shall be coupled together to form a 1C container and placed on four level supports, one under each extreme outer corner fitting, and shall then have a load uniformly distributed over the floor in such a way that the gross weight of the four QUADCONs and the test load is 44,800 lbs. (199,270 Newtons). The QUADCON assembly shall then be restrained by rigid anchor points, two on one end only. These anchors shall be through the bottom apertures of the bottom corner fittings. A force of 89,600 lbs. (398,540 Newtons) (twice the gross weight of the QUADCON assembly) shall be applied horizontally to the QUADCON assembly through the bottom apertures of the other two corner fittings, first towards, and then away from the anchor points. Upon completion of this test, each QUADCON, and the assembly, shall show neither permanent deformation nor abnormality which will render it unsuitable for use and the dimensional requirements of 4.7.11 shall then be satisfied.

4.7.10 Coast Guard Certification. The QUADCONs shall be tested for Coast Guard Certification. If there is any damage to the QUADCONs due to these tests, it will be the responsibility of the contractor to repair the damage or make any correction necessary for the QUADCONs to pass Coast Guard certification at his own expense. A certification of Coast Guard approval shall be furnished and Coast Guard certification plates shall be attached to each delivered QUADCON.

4.7.11 Dimensional. The length, width, height, and diagonals (K1 and K2) shall be measured and recorded for each QUADCON. During first article testing, the length, width, height, and diagonals (K1 and K2) shall be measured and recorded for the four QUADCONs when coupled together to form a 1C container. The requirements of 3.7 shall be satisfied.

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4.7.12 Weight. The weight of each QUADCON shall be taken. The requirements of 3.8 shall be satisfied.

4.7.13 Environmental tests.

4.7.13.1 Weatherproofness. A stream of water shall be applied on all exterior joints and seams of the QUADCON under test from a nozzle of 0.5 in. (12.7 mm) inside diameter at a pressure of about 1 bar (corresponding to a head of about 33 ft. (10 m)) of water on the upstream side of the nozzle. The nozzle shall be held at a distance of 5 ft. (1.5 mm) from the container under test, and the stream shall be traversed at a speed of 4 in/s (100 mm/s). Procedures involving the use of several nozzles are acceptable provided that each joint or seam is subjected to a water loading no less that that which would be given by a single nozzle. At the conclusion of this test, the QUADCON shall be examined and show no evidence of leakage at any location.

4.7.13.2 Humidity resistance. The QUADCON shall be subjected to MIL-STD-810, Method 507.1, Procedure II, excluding steps 2-4, 7, and 8. After cycling, the QUADCON shall be examined and show no evidence of delamination, cracking, corrosion, or deterioration of any part of the QUADCON.

4.7.13.3 Temperature range. The QUADCON shall be cold soaked in a mechanically refrigerated cold chamber at -70°F for a minimum of 16 hours. The chamber temperature shall then be raised to -65°F. The QUADCON doors will be opened and closed. Any malfunction shall be recorded and reported. The QUADCON shall then be visually examined both internally and externally for material degradation. There shall be no damage to seals or other components and all hardware shall operate as specified during and after completion of this test.

4.7.13.4 Temperature shock. A representative QUACON panel specimen measuring 24 in. by 48 in. (610 mm by 1,220 mm) shall be subjected to MIL-STD-810, Method 503.1. Upon completion of this test, there shall be no separation, cracks, delamination, or degradation of physical properties. Superficial cracks in the glazing which will not affect serviceability of the material are acceptable. The Government may require testing to confirm that the physical properties are not affected.

4.7.13.5 Roof load, uniform. The QUADCON under test shall be placed on the ground and leveled. The entire roof of the QUADCON shall then be uniformly loaded with a weight of 1,533 lbs. per square foot (73,392 N/m) and allowed to remain loaded for a period of 5 minutes. Upon completion of this test, the QUADCON shall show neither permanent deformation nor abnormality which will render it unsuitable for use and the dimensional requirements of 4.7.11 shall then be satisfied.

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4.7.13.6 Roof load, concentrated. A load of 660 lbs. (2,936 Newtons) shall be uniformly distributed over an area of 12 in. by 24 in. (305 mm by 610 mm) located at the weakest (nonsupported) area of the roof of the QUADCON. This load shall be allowed to remain there for a period of 5 minutes. Upon completion of this test, the QUADCON shall show neither permanent deformation nor abnormality which will render it unsuitable for use and the dimensional requirements of 4.7.11 shall then be satisfied.

4.7.13.7 Stacking. The four QUADCONs under test shall be coupled together to form a 1C container and placed on four level pads, one under each extreme outer corner fitting of the assembly. The pads shall be centered under the fittings and be substantially of the same plan dimensions as the fittings. The QUADCON assembly under test shall then be subjected to a vertical force equivalent to a load of 100,800 lbs. (448,358 Newtons), applied for 5 minutes simultaneously on each pair of upper end fittings in such a manner that the planes of application of the forces and the support of the QUADCON assembly remain horizontal and unchanged during the test. The forces shall be applied through a corner fitting or a pad of the same plan area as a corner fitting. Each pad shall be offset in the same direction by 1 in. (25.4 mm) laterally and 1.5 in. (38 mm) longitudinally. Upon completion of this test, each QUADCON, and the assembly, shall show no permanent deformation nor abnormality which will render it unsuitable for use and the dimensional requirements of 4.7.11 shall then be satisfied for each QUADCON individually and as an assembly.

4.7.13.8 Sand and dust. The QUADCON shall be subjected to MIL-STD-810, Method 510.1, Procedure I. There shall be no damage to seals or other components and all hardware shall operate as specified during and upon completion of this test.

4.7.13.9 Sunshine. Representative samples of all gaskets and seals, as well as a 12 in. by 12 in. (305 mm by 305 mm) panel sample, shall be subjected to MIL-STD-810, Method 505.1, Procedure I. Upon completion of this test, the samples shall be visually examined and any evidence of degradation of physical properties shall constitute failure of this test.

4.7.13.10 Solar load. The QUADCON shall be subjected to a simulated solar load sufficient to raise the outer skin temperature to a minimum of 200°F uniformly over the fixed roof. The uniform solar load temperature should be gradually attained within 4 hours and shall be maintained for an additional 4 hours. As a minimum, one thermocouple per 10 square feet (one square meter) shall be uniformly distributed on the entire area of the roof. All of the thermocouples should read 200°F ± 15°F throughout the 4 hour period that the solar load shall be maintained. Upon completion of this test, the roof panels shall be examined and any evidence of delamination or deformation shall constitute failure of this test.

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4.7.13.11 Marine atmosphere. One representative sample of each fastener, seal, panel, hinge, and corner fitting which will be exposed to the atmosphere in the operation and storage modes shall be finished in accordance with the applicable drawings and subjected to tests as described in MIL-STD-810, Method 509.1, except that the salt solution shall have a concentration of 10 percent and the exposure period shall be 96 hours. Upon completion of this test, the test items shall show no evidence of delamination, cracking, corrosion, or deterioration.

4.7.14 Shock tests

4.7.14.1 Drop shock. The QUADCON shall be uniformly loaded to the rate gross weight of 11,200 lbs. (49,840 Newtons) and lifted 12 in. (305 mm) above a flat concrete surface. The distance between the concrete surface and each lower corner fitting of the QUADCON under test shall not vary more than 0.5 in. (12.7 mm) at the time of the drop. The QUADCON shall then be allowed to free fall to the concrete surface. This test shall be done separately with the container rack and inserts inside if specified in the contract (see 3.6.10). Upon completion of this test, the QUADCON shall show neither permanent deformation nor abnormality which will render it unsuitable for use and the dimensional requirements of 4.7.11 shall then be satisfied.

4.7.14.2 Rotational drop. The QUADCON shall be uniformaly loaded to the rated gross weight of 11,200 lbs. (49,840 Newtons) with one edge supported 4 in. (102 mm) above a flat concrete surface. The opposite edge shall be lifted a distance of 12 in. (305 mm) above the concrete surface and allowed to free fall onto the surface. This procedure shall be repeated until each edge has been allowed to free fall. This test shall be done with the container rack and inserts inside if specified in the contract (see 3.6.10). Upon completion of this test, the QUADCON shall show neither permanent deformation nor abnormality which will render it unsuitable for use and the dimensional requirements of 4.7.11 shall then be satisfied.

4.17.15 Transportability.

4.17.15.1 Railroad transportability. The QUADCON shall be uniformly loaded to the rated gross weight of 11,200 lbs. (49,840 Newtons) and subjected to MIL-STD-810, Method 516.3, Procedure VIII, with impact speeds between 9 mph (14.5 km/hr) and 10 mph (16.1 km/hr) in both directions. The QUADCON shall be subjected to two impacts in each direction. This test shall be done with the QUADCON racks and inserts inside if specified in the contract (see 3.5.2). Upon completion of this test, the QUADCON shall show neither permanent deformation nor abnormality which will render it unsuitable for use and dimensional requirements of 4.7.11 shall then be satisfied.

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4.17.15.2 Air transportability. The QUADCON shall be subjected to MIL-STD-1791, paragraph 5.2.5., Rapid Decompression Verification, and shall meet the requirements for certification in accordance with MIL-STD-209.

4.8 Test reports. The results of all testing shall be reported and written in compliance with MIL-STD-831 fully describing and illustrating the test procedures and test results.

5. PACKAGING

5.1 Preservation. The QUADCON requires no additional preservation.

5.2 Packing. As the QUADCON is a 1F ISO container, no protective media need be installed prior to shipment. Conventional methods of blocking, bracing, and strapping are to be employed during shipment by rail, road, ship and air as described in MIL-STD-1186. When feasible, four QUADCONs shall be coupled together and shipped as a 1C ISO container.

5.3 Marking. In addition to the marking required by 3.10, each QUADCON shall be marked for shipment in accordance with the requirements of MIL-STD-129, paragraph 4.4.8. A removable data plate shall be installed to show cube, weight, and destination specified by the contracting activity.

6. NOTES

6.1 Intended use. The QUADCON covered by this specification is one of a family of containers intended for use by the Fleet Marine Force to house and support the storage and movement of organizational property and consumable supplies in the field and in garrison.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When first article testing is required (see 3.1).

c. When railroad transportability testing with the container rack and inserts inside is required (see 3.5.2).

d. When drop shock testing with the container rack with inserts inside is required (see 3.6.10).

e. When a quality control system is required (see 4.1.1).

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6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approve Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is no used, the data specified below shall be delivered by the contract in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Para No.	Data Requirements Title	Applicable DID No.	Option

(Data item descriptions related to this specification and identified in section 6 will be approved and listed as such in DOD 5000.19L, Vol. II, AMSDL. Copies of data item descriptions required by the Contractor in connection with specified acquisition functions should be obtained from the Naval Publications and Form Center or as directed by the Contracting Officer.)"

6.3 First article. First Article test items shall be four sample items manufactured using the same procedures and tooling proposed for production. The Contractor shall submit a First Article test plan and receive Government approval prior to testing. The First articles shall be examined and tested for approval at the Contractor's plant or at an independent laboratory acceptable to the Government. First Article tests shall be witnessed by a Government representative for the acquiring activity. After completion of First Article testing the Contractor shall submit a separate test report. The Government reserves the right to waive the requirement for First Article testing for items that have previously been tested and accepted by the Government.

6.4 Subject term (keyword) listing. The following subject term (key word) listing is furnished for retrival searches.

Container

Container, QUADRUPLE

Container, Rack, QUADCON

Locker, Small item, Storage

QUADCON

6.5 International standardization agreements. Certain provisions of this specification (see 3.1) are the subject of international standardization agreement ISO/DIS 668 and are regulated by the Coast Guard. When amendments, revision, or cancellation of the

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specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accommodations.

Preparing Activity
Navy-MC
Project No. 8145-0036

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-29445(MC)	Container, Quadruple (Quadcon)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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